In [1]:
```
#SLICING  OPERATIONS ON NDARRAY--1D Array
```

In [2]:
```
import numpy as np
```

In [3]:
```
lst=[10,20,30,40,50,60,70,80,90]
a=np.array(lst)
print(a,type(a))
```

```
[10 20 30 40 50 60 70 80 90] <class 'numpy.ndarray'>
```

In [4]:
```
a[2:5]
```

Out[4]:
```
array([30, 40, 50])
```

In [5]:
```
a[2:]
```

Out[5]:
```
array([30, 40, 50, 60, 70, 80, 90])
```

In [6]:
```
a[:5]
```

Out[6]:
```
array([10, 20, 30, 40, 50])
```

In [7]:
```
a[::2]
```

Out[7]:
```
array([10, 30, 50, 70, 90])
```

In [8]:
```
a[::-1]
```

Out[8]:
```
array([90, 80, 70, 60, 50, 40, 30, 20, 10])
```

In [9]:
```
#SLICING  OPERATIONS ON NDARRAY--2D Array
```

In [10]:
```
lst=[10,20,30,40,50,60,70,80,90]
a=np.array(lst)
a.shape=(3,3)
print(a,type(a))
```

```
[[10 20 30]
 [40 50 60]
 [70 80 90]] <class 'numpy.ndarray'>
```

In [11]:
```
a[0:3,1:2]
```

Out[11]:
```
array([[20],
       [50],
       [80]])
```

In [12]:
```
a[:,1:2]
```

Out[12]:
```
array([[20],
       [50],
       [80]])
```

In [13]:
```
a[::,1:2]
```

Out[13]:
```
array([[20],
       [50],
       [80]])
```

In [14]:
```
a[::1,1:2:1]
```

Out[14]:
```
array([[20],
       [50],
       [80]])
```

In [15]:
```
print(a)
```

```
[[10 20 30]
 [40 50 60]
 [70 80 90]]
```

In [16]:
```
a[1:3,0:3]
```

Out[16]:
```
array([[40, 50, 60],
       [70, 80, 90]])
```

In [17]:
```
a[1:,:]
```

Out[17]:
```
array([[40, 50, 60],
       [70, 80, 90]])
```

In [18]:
```
a[1::1,::]
```

Out[18]:
```
array([[40, 50, 60],
       [70, 80, 90]])
```

In [19]:
```
a[1::1,::1]
```

Out[19]:
```
array([[40, 50, 60],
       [70, 80, 90]])
```

In [20]:
```
print(a)
```

```
[[10 20 30]
 [40 50 60]
 [70 80 90]]
```

In [21]:
```
a[0:2,1:3]
```
Out[21]:
```
array([[20, 30],
       [50, 60]])
```

In [22]:
```
a[:2,1:]
```
Out[22]:
```
array([[20, 30],
       [50, 60]])
```

In [23]:
```
a[:2:1,1::]
```
Out[23]:
```
array([[20, 30],
       [50, 60]])
```

In [24]:
```
a[:2:1,1::1]
```
Out[24]:
```
array([[20, 30],
       [50, 60]])
```

In [25]:
```
print(a)
```
```
[[10 20 30]
 [40 50 60]
 [70 80 90]]
```

In [26]:
```
a[0:3:2,0:3]
```
Out[26]:
```
array([[10, 20, 30],
       [70, 80, 90]])
```

In [27]:
```
a[::2,:]
```
Out[27]:
```
array([[10, 20, 30],
       [70, 80, 90]])
```

In [28]:
```
a[::2,::]
```
Out[28]:
```
array([[10, 20, 30],
       [70, 80, 90]])
```

In [29]:
```
a[::2,::1]
```
Out[29]:
```
array([[10, 20, 30],
       [70, 80, 90]])
```

In [30]:
```
print(a)
```
```
[[10 20 30]
 [40 50 60]
 [70 80 90]]
```

In [31]:
```
a[0:3:2,0:3:2]
```

Out[31]:
```
array([[10, 30],
       [70, 90]])
```

In [32]:
```
a[::2,::2]
```

Out[32]:
```
array([[10, 30],
       [70, 90]])
```

In [33]:
```
a[0::2,0::2]
```

Out[33]:
```
array([[10, 30],
       [70, 90]])
```

In [34]:
```
a[:3:2,:3:2]
```

Out[34]:
```
array([[10, 30],
       [70, 90]])
```

In [36]:
```
#SLICING  OPERATIONS ON NDARRAY--nD Array
lst=[10,20,30,40,50,60,70,80,90,15,25,35,65,75,85,15,55,65]
a=np.array(lst)
a.shape=(3,2,3)
print(a,type(a))
```

```
[[[10 20 30]
  [40 50 60]]

 [[70 80 90]
  [15 25 35]]

 [[65 75 85]
  [15 55 65]]] <class 'numpy.ndarray'>
```

In [37]:
```
a[0:3,0:1,0:3]
```

Out[37]:
```
array([[[10, 20, 30]],

       [[70, 80, 90]],

       [[65, 75, 85]]])
```

In [38]:
```
print(a)
```

```
[[[10 20 30]
  [40 50 60]]

 [[70 80 90]
  [15 25 35]]

 [[65 75 85]
  [15 55 65]]]
```

In [39]:
```
a[0:2,:,1:3]
```

Out[39]:
```
array([[[20, 30],
        [50, 60]],

       [[80, 90],
        [25, 35]]])
```

In [40]:
```
a[0:2,::,1:]
```

Out[40]:
```
array([[[20, 30],
        [50, 60]],

       [[80, 90],
        [25, 35]]])
```

In [41]:
```
a[0:2:1,::1,1::1]
```

Out[41]:
```
array([[[20, 30],
        [50, 60]],

       [[80, 90],
        [25, 35]]])
```

In [42]:
```
print(a)
```

```
[[[10 20 30]
  [40 50 60]]

 [[70 80 90]
  [15 25 35]]

 [[65 75 85]
  [15 55 65]]]
```

In [43]:
```
a[0:3,0:1,::2]
```

Out[43]:
```
array([[[10, 30]],

       [[70, 90]],

       [[65, 85]]])
```

In [44]:
```
a[:,:1,::2]
```

Out[44]:
```
array([[[10, 30]],

       [[70, 90]],

       [[65, 85]]])
```

In [45]:
```
a[::1,:1:1,::2]
```

Out[45]:
```
array([[[10, 30]],

       [[70, 90]],

       [[65, 85]]])
```

In [ ]:
```
```